Problem 1

Cost analysis. A farm implements dealer is seeking a fourth warehouse location to complement three existing warehouses. There are three potential locations: Charlotte, N.C.; Atlanta, Ga.; and Columbia, S.C. Charlotte would involve a fixed cost of $4,000 per month and a variable cost of $4 per unit; Atlanta would involve a fixed cost of $3,500 per month and a variable cost of $5 per unit; and Columbia would involve a fixed cost of $5,000 per month and a variable cost of $6 per unit. Use of the Charlotte location would increase system transportation costs by $19,000 per month, Atlanta by $22,000 per month, and Columbia by $18,000 per month. Which location would result in the lowest total cost to handle 800 units per month?

Solution

Given: Volume = 800 units per month

	FC per Month	Variable Cost per Unit, v	Transportation Cost per Month
Charlotte	$4,000	$4	$19,000
Atlanta	3,500	5	22,000
Columbia	5,000	6	18,000

Monthly total cost = FC + VC + Transportation cost

Charlotte: $4,000 + $4 per unit × 800 units + $19,000 = $26,200

Atlanta: 3,500 + 5 per unit × 800 units + 22,000 = 29,500

Columbia: 5,000 + 6 per unit × 800 units + 18,000 = 27,800

Hence, Charlotte would have the lowest total cost for this monthly volume.

Problem 2

Profit analysis. A manufacturer of staplers is about to lose its lease, so it must move to another location. Two sites are currently under consideration. Fixed costs would be $8,000 per month at site A and $9,400 per month at site B. Variable costs are expected to be $5 per unit at site A and $4 per unit at site B. Monthly demand has been steady at 8,800 units for the last several years and is not expected to deviate from that amount in the foreseeable future. Assume staplers sell for $6 per unit. Determine which location would yield the higher profit under these conditions.

Solution

Profit = $Q(R - v) - FC$

Site	Revenue	FC	v	Monthly Profit
A	$52,800	$8,000	$44,000	$ 800
B	$52,800	$9,400	$35,200	$8,200

Hence, site B is expected to yield the higher monthly profit.

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Problem 3

Factor rating. Determine which location has the highest factor rating given the following information:

Factor	Weight	LOCATION A	LOCATION B
Labor cost	.50	20	40
Material cost	.30	10	30
Supplier base	.20	50	10
	1.00		

Solution

Combining the weights with the location scores, we can see that location B has the highest score:

Factor	Weight	LOCATION A	LOCATION B	WEIGHTED SCORES A	WEIGHTED SCORES B
Labor cost	.50	20	40	.50(20) = 10	.50(40) = 20
Material cost	.30	10	30	.30(10) = 3	.30(30) = 9
Supplier base	.20	50	10	.20(50) = 10	.20(10) = 2
	1.00			23	31

Problem 4

Center of gravity. Determine the center of gravity location for these destinations:

Destination	*x,y* Coordinates	Weekly Quantity
D1	3,5	20
D2	6,8	10
D3	2,7	15
D4	4,5	15
		60

Solution

If the weekly quantities had all been equal, we could have used the two formulas in Equation to find the center of gravity. Because the weekly quantities are not all equal, we must use the formulas in Equation 8–4.

$$\bar{x} = \frac{\Sigma xQ}{\Sigma Q} = \frac{3(20) + 6(10) + 2(15) + 4(15)}{60} = \frac{210}{60} = 3.5$$

$$\bar{y} = \frac{\Sigma yQ}{\Sigma Q} = \frac{5(20) + 8(10) + 7(15) + 5(15)}{60} = \frac{360}{60} = 6.0$$

Hence, the center of gravity has the coordinates $x = 3.5$ and $y = 6$.

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